

Data Analysis for Quantum Vehicle Routing Problem (VRP) Research

This document outlines a comprehensive data analysis plan for evaluating the performance of a quantum algorithm, specifically the Quantum Approximate Optimization Algorithm (QAOA), on the Vehicle Routing Problem (VRP). The analysis is designed to benchmark the quantum solver against classical solutions and to provide critical insights into its efficiency, accuracy, and resilience to noise. The plan is organized into seven key calculations, each providing a unique perspective on the algorithm's performance.

1. Optimality Ratio

The optimality ratio is the primary calculation for assessing the solution quality of a quantum approximation algorithm. It measures how closely the average solution from the quantum computer approximates the true optimal solution.

- **Calculation:** The ratio is calculated as the true minimum cost found by the classical solver (C_{min}) divided by the expected cost ($\langle C \rangle$) of the quantum solution. The expected cost is derived from the full probability distribution of quantum measurements.

$$\text{Optimality Ratio} = \frac{C_{min}}{\langle C \rangle}$$

- **Insights:** A ratio closer to 1 indicates a high-quality solution. This calculation is fundamental for **benchmarking** and allows for a direct comparison of the quantum algorithm's performance against the best possible classical solution. It helps to answer the question: "How good are the solutions produced by our quantum algorithm?"

2. Invalid Bitstring Measurement

Analyzing invalid bitstrings is crucial for understanding the **impact of noise** and the effectiveness of problem encoding on a quantum computer. Invalid bitstrings are measurements that correspond to solutions that violate the problem's constraints (e.g., a path that doesn't visit all nodes).

- **Calculation:** The number of invalid measurements is the difference between the total shots (total measurements) and the sum of counts for all valid bitstrings recorded.

$$N_{\text{invalid}} = N_{\text{shots}} - \sum_i \text{count}_i$$

- **Insights:** A high number of invalid measurements can be an indicator of significant **noise** in the quantum processor, which can cause the measured state to decohere into an unphysical solution. This analysis can also test if the penalty terms in the QAOA problem formulation are strong enough to suppress invalid solutions.

3. Comparison of ‘nodevisittime’ with ‘naiveadjacency’

This comparison directly addresses the **trade-off between resource efficiency (qubits) and solution accuracy**. Different problem encodings can use a varying number of qubits to represent the same problem.

- **Calculation:** Compare the optimality ratio and resource usage (e.g., number of qubits) of the ‘nodevisittime’ encoding against the ‘naiveadjacency’ encoding for the same problem instance. The ‘nodevisittime’ encoding uses fewer qubits for the same problem size.
- **Insights:** This analysis helps determine the practical utility of different encodings on current, noisy quantum hardware. It can demonstrate whether the gains from a more qubit-efficient encoding (i.e., less noise) outweigh any potential loss in solution quality compared to a less efficient but potentially more accurate encoding.

4. Comparison Between Shots

This analysis explores the impact of **quantum sampling statistics** on the final solution quality. Quantum computers are probabilistic, and each run generates a probability distribution of possible answers.

- **Calculation:** Compare the optimality ratio and the spread of measured costs (e.g., standard deviation) for runs with different numbers of shots.
- **Insights:** In theory, a larger number of shots should lead to a more stable and accurate estimate of the expected cost, as it reduces the statistical uncertainty in your measurements. This analysis can reveal if there is a point of diminishing returns where increasing the number of shots no longer significantly improves the result due to dominant systematic noise.

5. Comparison of QAOA Depth

QAOA depth is a proxy for the algorithm’s **computational power** and **complexity**. A deeper circuit involves more operations and can, in principle, explore a larger solution space.

- **Calculation:** Compare the optimality ratio and the number of invalid bitstrings across different QAOA depths.
- **Insights:** This is a crucial analysis for any QAOA study. A deeper circuit might find better solutions, but it also accumulates more noise. This analysis helps determine the **optimal trade-off** between the algorithm’s expressiveness and the detrimental effects of noise. It can show if there is a “sweet spot” depth that yields the best performance on a given quantum processor.

6. Cost Distribution Analysis

Instead of just relying on the average cost, analyzing the full **cost probability distribution** provides a richer understanding of how the algorithm is exploring the solution space.

- **Calculation:** Visualize the probability distribution of measured costs in a histogram or density plot.
- **Insights:** This analysis provides insights beyond a single number. You can observe if the distribution is shifting towards the true minimum as you increase the QAOA depth, indicating **convergence**. A broad or multimodal distribution may indicate high noise or that the algorithm is getting “stuck” in local minima. A large spread can also reveal the inherent uncertainty of the probabilistic quantum solution.

7. Scaling Analysis

This is arguably the most important analysis for any quantum algorithm study as it directly addresses the question of **quantum advantage**. It evaluates how the algorithm’s performance changes as the problem size increases.

- **Calculation:** Plot a performance calculation, such as the optimality ratio, against the problem size (e.g., ‘num_nodes’). Separate curves can be plotted for different configurations, such as the ‘nodevisittime’ and ‘naiveadjacency’ encodings.
- **Insights:** This analysis shows if the quantum algorithm’s performance degrades gracefully or precipitously as the problem becomes more complex. It can also be used to identify a potential **crossover point** where

the quantum algorithm might begin to outperform classical algorithms as the problem scales up, providing evidence for the potential of quantum advantage.

Limitations and Future Works

- The current analysis is limited by the number of problem instances and the range of QAOA parameters tested.
- The results are based on a specific quantum hardware, and performance may vary on different platforms.
- Future work should include analyzing other problem types within VRP, such as those with time windows or multiple depots.
- Further research can also explore more sophisticated quantum-classical hybrid approaches to parameter optimization.
- The analysis could be extended to include a more detailed characterization of the noise present in the quantum device.
- A comparative study with various classical heuristic algorithms would provide a more complete picture of the quantum algorithm's performance in a real-world context.
- Future research can focus on developing advanced error mitigation techniques tailored to the VRP problem.
- Exploring the use of more powerful quantum architectures, like universal gate-based quantum computers, may yield improved results.